

Integrating Structured Knowledge with Machine and Deep Learning for Fake News Detection

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Abstract — Knowledge graphs (KGs) are key structural elements in the organization of structured information and are applicable to tasks such as semantic search, recommendation or misinformation detection, among others. In this work, we explore two such complementary paradigms characterizing KG construction and utilization: (1) classic machine learning (ML) techniques for building KGs (e.g., Support Vector Machines (SVMs) for entity recognition, Random Forests for relation extraction), and (2) deep learning (DL) approaches (e.g., BERT/Roberta transformers for text encoding, graph neural networks (GNNs)) for leveraging KGs in fake news detection. This survey evaluates the machine learning (ML) approaches for entity recognition, relation extraction, and knowledge representation, with a specific focus on their interpretability and scalability. We surveyed deep learning methods for fake news detection, focusing on models (BERT, Roberta) for contextual text analysis, graph neural networks (GNNs) and hybrid models combining KGs with sequential models. We found that while ML is a more common and reliable approach to KG construction, DL enables more complex pattern extraction for tasks such as misinformation detection.

Keyword *Fake — news detection, Knowledge Graphs (KGs), machine learning, Deep Learning, Transformers, Graph neural networks*

I. INTRODUCTION

As social media has become a dominant way to learn and share news since 2015, fake news has spread like wildfires in the digital age. Public health emergencies, broken trust in institutions, and social divisions are consequences of this global phenomenon, that disseminates misinformation at an unprecedented speed. Two notable examples are the spread of misinformation during the 2020 U.S. presidential elections that undermined public trust in elections [1,2], and the COVID-19 infodemic, where misinformation about vaccines and

treatments has led to vaccine hesitancy and slowed the pandemic response [3,4].

Misinformation in this modern era is both complex and deeply contextualized; a unique challenge to counter and combat in one of the most complex and nuanced narratives in recent history. The article distinguishes misinformation (false or misleading content shared without intent to deceive) from disinformation (deliberately fabricated and propagated falsehoods), while fake news encompasses both, often with sensationalized or manipulative narratives [5,6]. These definitions remain consistent throughout the discussion, particularly in analyzing election interference (disinformation) and health-related false claims (misinformation).

Research has shown that misinformation spreads faster than true content, even because of emotional narratives [5]. In addition, the advent of deception methods from deepfakes to fabricated news and manipulative narrative tactics has blurred the line between what is real and what is fake [6]. Existing methods, although Machine Learning and Natural Language Processing techniques show promise in fake news detection, tend to miss the need for more overarching and context-dependent common sense [7].

Knowledge graphs (KGs) [8], which are structured representations of entities and their relationships, provide substantial improvement over misinformation detection in both classical supervised and deep-learning methods. This amalgamation ensures a robust information framework that utilizes the strength of every paradigm: ML provides interpretable feature engineering and scalable knowledge graph creation, and DL provides the ability to recognize complex patterns and put them into context [9,10]. This hybrid supports normalized information being analyzed at scale by the ML-driven population of the knowledge graph through entity and relation extraction, complex semantic reasoning through DL-powered graph neural networks, and claims cross-verified

against structured knowledge bases [11]. For example, ML methods can stably extract factual relations from heterogeneous sources to assist in the construction and population of KGs, allowing DL models to study these graphs to identify subtle inconsistencies, trends in novel forms of misinformation, or changing narratives that rule-based systems may not capture [12]. According to recent research, these integrated systems achieve better results than static isolated approaches by leveraging the explainability of knowledge-awareness in a symbolic representation and the representational power of neural networks [13]. The interplay of knowledge graphs, machine learning, and deep learning ideas provides a powerful answer to the developing difficulties of misinformation at scale, as discussed in the following sections.

II. RESEARCH METHODOLOGY

This section defines the key research questions guiding our study, outlines the systematic search process, and delineates the scope of our investigation into KG-based strategies for fake news detection using ML and DL techniques.

A. Research Questions

To evaluate the role of structured knowledge in fake news detection, we formulated the following research questions:

- RQ1: How do knowledge graphs enhance fake news detection compared to unstructured data approaches?
- RQ2: What ML techniques (e.g., SVMs, Random Forests) are most effective for KG construction (entity/relation extraction)?
- RQ3: What DL architectures (e.g., GNNs, transformers, hybrid models) best leverage KGs for misinformation classification?
- RQ3: How do KG embedding methods (e.g., TransE, RotatE, ComplEx) improve relational reasoning in fake news detection?

These questions guide our comparative analysis of KG-driven fake news detection frameworks.

B. Search Process

We conducted a systematic literature review using Google Scholar, IEEE Xplore, and ACM Digital Library, focusing on peer-reviewed papers (2018–2024) with the following keywords:

- **Knowledge Graphs:** ("Knowledge Graph" OR "KG Embeddings" OR "Graph Neural Networks").
- **ML/DL Models:** ("Machine Learning" OR "Deep Learning" OR "Transformers" OR "BERT" OR "RoBERTa").
- **Application Domain:** ("Fake News Detection" OR "Misinformation" OR "Disinformation" OR "Hoax Detection").

The selection of the 21 high-impact papers was based on four key criteria: relevance to KG-based fake news detection using ML/DL approaches, impact measured through publication venue prestige and citation counts, methodological rigor

including reproducibility and clear evaluation metrics, and temporal coverage limited to 2018–2024 to ensure contemporary relevance. Potential biases in this selection process include venue bias favoring certain conferences/journals, citation bias toward established works over newer methods, and language bias from excluding non-English publications. These limitations may influence the generalizability of our conclusions, particularly regarding emerging techniques or region-specific approaches.

III. INTEGRATING KNOWLEDGE GRAPHS WITH MACHINE LEARNING FOR FAKE NEWS DETECTION

For improving the fake news detection, machine learning models are now focusing on the knowledge graph embeddings. Through such method, structured entities and relations can be translated into vector space, that models can hence drive their performances upon a deeper semantic understanding. The next section discusses representative embedding approaches empowering these developments.

A. Supervised Learning for Entity and Relation Extraction

The conversion process of unstructured news contents into structured knowledge graphs are performed using supervised machine learning techniques (fig 1). In the field of named entity recognition, models such as Conditional Random Fields (CRF equation 1), and Support Vector Machines (SVM) have shown a tendency of achieving better results [6], by using well-annotated training data to achieve efficiency metrics (F₁-scores 0.79 for relation detection and F₁-scores 0.77 for relation type classification) [16]. Such models are highly effective in recognizing and classifying important entities (people, organizations, locations, etc.) mentioned in news content and the semantic relationships between them. Ensemble methods such as Random Forests and logistic regression models for relation extraction tasks learn to identify the type of relation (such as "claims," "denies," "supports") by analyzing patterns of linguistic and syntactic features, and contextual clues. These structured triples (subject-predicate-object) result in basic units which can then be used to create large-scale knowledge graphs for further verification and analysis [15].

$$P(Y|X) = \frac{1}{Z(X)} \exp \left(\sum_i \sum_k \lambda_k f_k(y_{i-1}, y_i, X, i) \right) \quad (1)$$

$P(Y|X)$: Probability of label sequence Y given input X .

- $Z(X)$: Normalization factor.
- λ_k : Learned weights for feature functions f_k .
- f_k : Feature functions capturing transitions and emissions.

Once the entities and their relations are extracted, the next step is to represent them in a way that allows the processing of deeper semantic knowledge. This is where knowledge graph embeddings bridge: they offer a scalable way to perform effective fake news detection in a vectorial reasoning space.

B. Knowledge Graph Embedding Techniques for Fake News Detection

Machine learning techniques that represent entities and relations as low-dimensional dense vector embeddings have revolutionized the field of knowledge representation, providing a means for efficient reasoning and link prediction. There are three main embedding methods that have been proven to work well for fake news detection.

At the core, the basic Translating Embeddings (TransE) model of relationships, a paragon of simplicity and strength [17], conceives relations as vector offsets between entities. This means that for each triplet (h, r, t) , where h is the head entity, r is the relation, and t is the tail entity, the embedding of the tail entity approximately equals the sum of the embedding of head entity and relation ($h + r \approx t$) which yields a good solution (78–82% accuracy in link prediction tasks [19]) with a relatively low computational cost. Although TransE is still applied to fake news detection, using these embeddings to confirm entity relations between news articles, the performance achieves an F1-score of 89.17% on the SLN dataset. The DEAP-FAKED framework [25] also demonstrates that TransE embeddings can be combined into performance-oriented vectorized textual features and well-structured knowledge built-out from Wikidata can improve the classification accuracy from the baseline level to a score of 90% in fake news classification tasks.

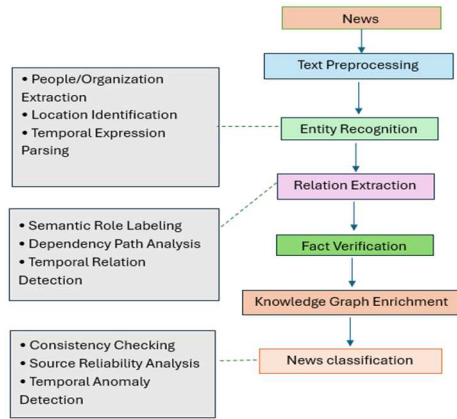


Figure 1: Text processing pipeline for fake news detection, integrating entity extraction, relation analysis, and knowledge graph verification.

RotatE [20] applies a similar translational idea, but in a complex vector space, enabling better modeling of symmetric, antisymmetric, and inverse relations by representing relations as rotations. In another study [21] that focused on heterogeneous representation ensembles, the authors combined RotatE embeddings with RoBERTa text features and reported a state-of-art performance including a remarkable 95.7% accuracy on COVID-19 misinformation detection. In the multi-domain fake news, the KG-MFEND model [22] applied RotatE to leverage those rotational properties in keeping the representational consistency across different topical domains and reached 87.63% accuracy.

ComplEx [23,24] works in fully complex space to capture asymmetric and many-to-many relationships and is capable of modeling the most complex relational patterns. The DEAP-

FAKED framework [25] uses complex embeddings (ComplEx) to describe semantic relationships between entities in news titles, which may also help explain its strong performance of 0.89%. DETERRENT [26] aims to verify medical claims are true or false by utilizing the same complex embeddings for healthcare misinformation detection, achieving an accuracy of 92.06%, especially for difficult cases with bidirectional relationships.

Embedding techniques such as these are the building blocks of state-of-the-art knowledge graph fake news detection systems. As recent hybrid approaches [21] suggest, these strategic combinations of the two are often required to obtain optimal performance. An example of this is the KAPALM model [27], which combines several types of knowledge representations and achieves 91.34% accuracy on PolitFact data.

Unlike traditional machine learning algorithms that rely solely on surface-level linguistic patterns, knowledge graph-based techniques enhance misinformation detection by modeling deeper semantic and relational structures across news content. Empirical evidence—such as from the DEAP-FAKED [25] and KG-MFEND [22] frameworks demonstrate measurable performance improvements, including up to 5–10% increases in F1-score or accuracy, when knowledge graph embeddings are integrated. This suggests that knowledge graph techniques provide an additional layer of verification and relational reasoning that strengthens the reliability of fake news detection, beyond what standard machine learning models can achieve.

IV. INTEGRATING KNOWLEDGE GRAPHS WITH DEEP LEARNING FOR FAKE NEWS DETECTION

Although embedding methods have provided powerful means for capturing knowledge graph structure, recent progresses in deep learning have reshaped the way of using knowledge graphs. Leveraging text semantics, temporal sequences, and relation reasoning, deep learning approaches (such as combinations of LSTMs, GNNs, and Transformers) enhance the accuracy and scalability of fake news detection. The next section examines these main approaches and their role in the emerging picture of the study of misinformation.

A. Core Deep Learning Techniques for Knowledge Graph-Based Fake News Detection

Knowledge graphs-based fake news detection fundamentally consists of three intertwined deep learning approaches, each of which address with a different aspect of information verification fig 2.

- **Text Encoding:** LSTMs are mainly used to represent text information for example news titles or articles in vector format [30]. The biLSTM (bidirectional LSTM equation 2) in the DEAP-FAKED framework is designed to capture sequential information in news headlines. It then concatenates the textual representation with embeddings for the entities from a knowledge graph allowing the model to leverage both content and entity relations when faking news detection [19].

$$h_t = \text{BiLSTM}(x_t, h_{t-1}, h_{t+1}) \quad (2)$$

- x_t : Input vector at time step t
- h_{t-1} : Hidden state from the forward LSTM at time $t-1$
- h_{t+1} : Hidden state from the forward LSTM at time $t+1$

• **Entity representation learning:** The critical problem of disambiguation of entities that could have multiple meanings in various senses is solved in studies lead by A. Hogan et [28] and F. Scarselli et al. [31]. For instance, Washington could represent a person, state, or city, and word embeddings work badly for this kind of polysemy. To overcome this, more recent approaches build contextualized representations and encoding: (1) the importance of the entity in the knowledge graph topology, (2) the textual context from reputable text sources, and (3) the surrounding relationship in the actual news article being studied. L. Hu et al. [19] work achieved an accuracy of 91% due to attention mechanisms (equation 3) that automatically change their weights according to the verification task geographical relationships are given larger weights while analyzing location-based claims and professional relationships are given larger weights while analyzing claims about public figures.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V \quad (3)$$

- Q, K, V : Query, key, value matrices from Transformer (e.g., RoBERTa).
- d_k : Dimension of keys (scaling factor for stability).

• **Sequential processing techniques:** These methods tackle the time component of misinformation, as claims may not only conflict with known truths but can also change. Bidirectional LSTMs [30] with hierarchical attention explore not only the immediate context of a claim, but also its historical consistency in the knowledge graph. For example, a claim regarding "present inflation rates" would be fact-checked not only with the latest trend information in the knowledge graph however likewise within the general financial scenario. This enables

92% accuracy in temporal pattern recognition owing to the updates to long-term memory cells in the DEAP-FAKED model [25]. Temporal graph networks take this a step further by explicitly modelling changes in the relationships between entities over time, so that the system identifies only when a statement that was previously known or true (where we might have a relation like Person X holds position Y) is no longer true because of events that have since happened [27].

These deep learning methods serve as the preliminary work that interprets the context of content and entity. Furthermore, in order to better detection the fake news, it's essential to learn the structural relationship of entities through Graph Neural Network.

B. Graph Neural Networks (GNNs) for Fake News Detection

The complicated nature of all relationships within the fake news situation requires the structural reasoning abilities offered by Graph Neural Networks. One of the fundamental innovations of Graph Neural Networks (GNNs), as described by F. Scarselli et al. [31], is their base architecture for message-passing where the knowledge graph is propagated in a manner that brings out the aspect of logical reasoning of human beings and hence can be considered an innovative molecule of sorts. In this manner in the conventional fake news detection pipeline, the system first creates a subgraph containing all entities involved in the news item and its first-degree connections from the knowledge base. As an example, a political assertion may create a subgraph of politicians, organizations, locations, and events associated with the assertion.

Graph Attention Networks (GATs): P. Veličković et al. [29] enhanced this process and showed how trainable attention weights can be used to indicate the amount of information is allowed to flow between connected nodes. This enables the KAPALM model [27] to automatically reason that in a story discussing election interference, the relationship between a candidate and a foreign government should be more salient than their relationship to a local bakery, even though both relationships exist in the knowledge graph. These weights are then computed by an attention mechanism, which is dynamically calculated according to the node features and the verification task.

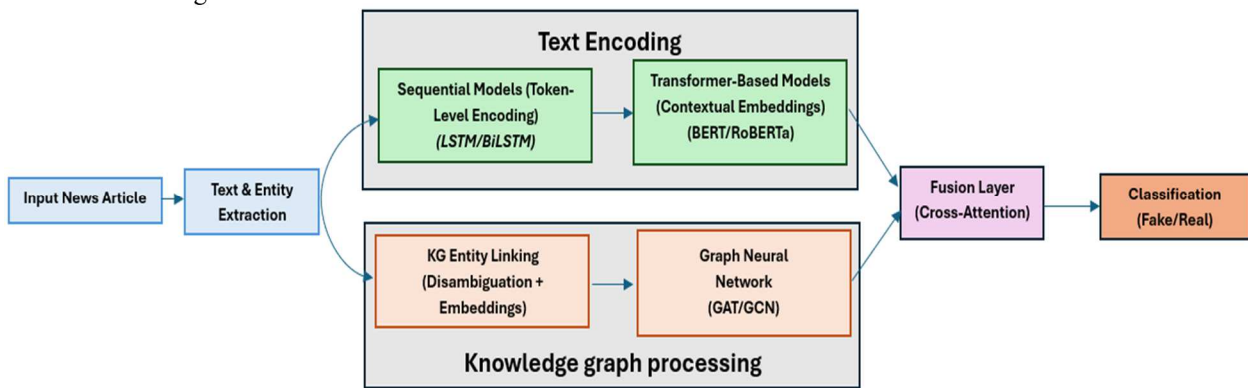


Figure 2: fake news classification integrating deep learning approaches and knowledge graph verification.

Graph convolutional networks (GCNs) (T. N. Kipf and M. Welling [17, 18]) take a different approach by aggregating features from a node's local neighborhood. When processing a claim that "celebrity X endorses product Y," the GCN may discover that although the endorsement relationship exists, it is surrounded by nodes indicating that the endorsement was satirical or expired, which would be missed by simpler approaches. The multi-layer architecture of GCNs achieves this by progressively combining information from first-order neighbors (direct connections) to higher-order neighbors (connections of connections), thereby capturing complex indirect relationships that often reveal the truth behind misleading claims. In CompareNet [19], this proves particularly useful for detecting fake news that relies on context-free truths.

GNNs become particularly evident when analyzing coordinated disinformation campaigns (J. Zhou et al. [32]). Traditional methods can flag individual claims as suspicious, however GNNs can detect trends across multiple articles by observing unusual clustering of entities, anomalous relational patterns, or sudden changes in connection weights that suggest artificial amplification. This structural analysis capability explains why GNN-based systems consistently outperform traditional methods by significant margins (X. Zhou and R. Zafarani [6], S. Qian et al. [14]), especially for sophisticated fake news that relies on carefully constructed half-truths and misleading associations.

Extending the structural reasoning power of GNNs the transformer models additionally improve the fake news detection task by bringing in strong language understanding with contextualized attention mechanisms.

C. Transformer Models for Enhanced Detection

Transformer architectures identified of fake news to the next level in the last decade or so, enabling the identification of sophisticated ways in which language is being used. Models such as BERT [24] and RoBERTa [18] are the state-of-the-art due to their bidirectional context processing, which is based on a fundamental innovation that allows for examining all words in a text in relation to each other all at once rather than sequentially, as was done earlier with RNNs and similar architectures. This becomes even more critical for recognizing advanced fake news techniques such as:

- Contextual manipulation: Providing a true fact but in misleading terms.
- Omission deception: The essential context is missing.
- Semantic attacks: Words used in an unusual and misleading manner.

Transformers integrated with Knowledge Graphs construct models thus become grounded in factual relationships rather than simply linguistic ones [28]. For example, the DEAP-FAKED model shows this hybrid setting where RoBERTa handles textual content while reading in the same time knowledge graph embeddings besitzten [25]. This helps the model spot contradictions that would otherwise fall under the

radar of text-only analysis for instance, if a news piece discusses a corporate executive making a political decision that does not match their position in the organizational hierarchy in the knowledge graph. self-supervised pre-training used by transformers (J. Devlin et al. [24], Y. Liu et al. [18]) gives them another superior quality for detecting fake news: the ability to learn from vast amounts of unlabeled data. This is particularly valuable because:

- The models are allowed to learn the statistics of true reporting before being fine-tuned on smaller label datasets.
- enables the models to create strong representations, which generalize well to unseen types of misinformation.
- handle the changing usage in real-time misinformation efforts.

An example of its application is CompareNet [19]. After extracting semantic meanings from news text with BERT embeddings, the model assimilates knowledge graph information using Graph Convolutional Networks (GCNs). Attention mechanisms fuse these two data streams by identifying which link and linked text are most relevant for verification at that moment in time. Transformers based model [19,20] being adaptive to real world fake news detection. They have the ability to process news in parallel, allowing them to analyze massive amounts of news content in a short time. And they also incorporate knowledge graphs to help them verify information against established facts.

To ensure knowledge graphs remain accurate and reliable while being updated in real time, systems typically implement layered update pipelines involving trusted data sources, temporal versioning, and human-in-the-loop verification. Many frameworks, such as DEAP-FAKED [25], also employ automated consistency checks and redundancy resolution strategies to maintain reliability. While real-time updates are critical to tracking fast-evolving misinformation, current literature acknowledges potential delays due to the trade-off between update speed and validation quality. The Ongoing works have to continue to explore scalable, real-time update mechanisms that preserve the integrity of knowledge graphs under dynamic conditions.

V. CONCLUSION

The integration of knowledge graphs (KGs) with AI techniques has emerged as a powerful strategy for combating misinformation. Transformer models like BERT and RoBERTa enhance contextual understanding and relational reasoning, while deep learning (DL) excels at uncovering complex patterns in fake news through semantic and structural analysis. However, traditional machine learning (ML) remains indispensable for constructing reliable KGs, offering interpretability and scalability in tasks such as entity and relation extraction. The most effective solutions combine structured knowledge representation with graph neural networks (GNNs), yet challenges persist in developing adaptive, multilingual, and explainable systems that can keep pace with the evolving nature of misinformation.

Future research should focus on advancing real-time KG updates to maintain accuracy in dynamic environments, with particular emphasis on attention mechanisms that can dynamically prioritize the most relevant knowledge elements for rapid verification. Improving cross-domain and multilingual generalization through adaptive attention architectures, and enhancing model explainability via interpretable attention weights will be crucial for fostering trust in automated detection. Hybrid architectures that merge ML's robustness in KG construction with attention-enhanced DL pattern recognition capabilities show particular promise, as demonstrated by frameworks like DEAP-FAKED and KAPALM, which achieve accuracy rates above 90%. Quantitative comparisons reveal that while ML methods achieve strong performance in structured tasks (e.g., F1-scores of 0.79 for relation extraction), attention-powered DL techniques outperform in detecting nuanced misinformation, with models like KG-MFEND reaching 87.63% accuracy. These hybrid approaches, leveraging attention-guided embeddings such as TransE and RotatE, highlight the synergy between symbolic knowledge and neural networks.

Ultimately, the fight against misinformation requires continuous innovation in scalable, adaptable, and transparent systems. By refining these technologies particularly through advanced attention mechanisms for real-time processing and addressing current limitations, future advancements can better safeguard the integrity of information in an increasingly complex digital landscape.

REFERENCES

- [1] B. Nyhan and J. Reifler, "When corrections fail: The persistence of political misperceptions," *Political Behavior*, vol. 32, no. 2, pp. 303-330, Jun. 2010.
- [2] P. J. Lavrakas, M. W. Traugott, C. Kennedy, B. L. Holbrook, and E. D. de Leeuw, "The future of U.S. general population telephone survey research," *Public Opinion Quarterly*, vol. 83, no. S1, pp. 289-308, 2019.
- [3] World Health Organization, "Managing the COVID-19 infodemic: Promoting healthy behaviours and mitigating the harm from misinformation and disinformation," Sep. 2020.
- [4] H. J. Larson, L. Z. Schulz, W. S. Tucker, and N. Smith, "The vaccine hesitancy scale: Psychometric properties and validation," *Vaccine*, vol. 36, no. 5, pp. 660-667, Jan. 2018.
- [5] S. Vosoughi, D. Roy, and S. Aral, "The spread of true and false news online," *Science*, vol. 359, no. 6380, pp. 1146-1151, Mar. 2018.
- [6] X. Zhou and R. Zafarani, "A survey of fake news: Fundamental theories, detection methods, and opportunities," *ACM Computing Surveys*, vol. 53, no. 5, pp. 1-40, Sep. 2020.
- [7] N. Ruchansky, S. Seo, and Y. Liu, "CSI: A hybrid deep model for fake news detection," in *Proc. ACM CIKM*, 2017, pp. 797-806.
- [8] A. Hogan, E. Blomqvist, M. Cochez, C. d'Amato, G. de Melo, C. Gutierrez, J. E. L. Gayo, S. Kirrane, S. Neumaier, A. Polleres, R. Navigli, A.-C. Ngonga Ngomo, S. M. Rashid, A. Rula, L. Schmelzeisen, J. F. Sequeda, S. Staab, and A. Zimmermann, "Knowledge graphs," *ACM Computing Surveys*, vol. 54, no. 4, pp. 1-37, Jul. 2021.
- [9] Y. Zhang, R. Zhang, Z. Zhang, and J. Li, "Benchmarking knowledge-enhanced NLP models," in *Proc. ACL*, 2021, pp. 322-335.
- [10] X. Wang, T. Gao, Z. Zhu, Z. Liu, J. Li, and J. Tang, "KG-BERT: BERT for knowledge graph completion," *arXiv:1909.03193*, 2019.
- [11] Z. Sun, W. Hu, Q. Zhang, and Y. Qu, "Knowledge graph alignment with entity-aware GNNs," in *Proc. WWW*, 2020, pp. 313-324.
- [12] W. Liu, H. Li, M. Yang, and X. Wang, "Hybrid reasoning for misinformation detection," in *Proc. AAAI*, 2022, pp. 12189-12197.
- [13] S. Ji, S. Pan, E. Cambria, P. Martinen, and P. S. Yu, "Survey on knowledge graph-enhanced NLP," *AI Open*, vol. 2, pp. 1-13, 2021.
- [14] S. Qian, J. Wang, J. Hu, Q. Fang, and C. Xu, "Knowledge-aware multi-modal adaptive graph convolutional networks for fake news detection," *ACM Transactions on Multimedia Computing, Communications, and Applications*, vol. 17, no. 3, pp. 1-23, 2021.
- [15] R. Mihalcea and C. Strapparava, "The lie detector: Explorations in the automatic recognition of deceptive language," in *Proc. ACL*, 2009, pp. 309-312.
- [16] Association for the Advancement of Artificial Intelligence, "End-to-end argumentation knowledge graph construction," Palo Alto, CA, USA, 2020.
- [17] A. Bordes, N. Usunier, A. Garcia-Duran, J. Weston, and O. Yakhnenko, "Translating embeddings for modeling multi-relational data," in *Proc. NIPS*, 2013, pp. 2787-2795.
- [18] Y. Liu, M. Ott, N. Goyal, J. Du, M. Joshi, D. Chen, O. Levy, M. Lewis, L. Zettlemoyer, and V. Stoyanov, "RoBERTa: A robustly optimized BERT pretraining approach," *arXiv:1907.11692*, 2019.
- [19] L. Hu, J. Yang, T. Wang, Y. Gao, and L. Nie, "Compare to the knowledge: Graph neural fake news detection with external knowledge," in *Proc. ACL*, 2021.
- [20] Z. Sun, Z.-H. Deng, J.-Y. Nie, and J. Tang, "RotatE: Knowledge graph embedding by relational rotation in complex space," in *Proc. ICLR*, 2019.
- [21] B. Koloski, T. Škrlić, S. Pollak, and N. Lavrač, "Knowledge graph informed fake news classification via heterogeneous representation ensembles," *Neurocomputing*, vol. 496, pp. 208-226, 2022.
- [22] L. Fu, X. Zhang, Y. Zhang, and W. Wang, "KG-MFEND: An efficient knowledge graph-based model for multi-domain fake news detection," *Journal of Supercomputing*, vol. 79, pp. 18417-18444, 2023.
- [23] T. Trouillon, J. Welbl, S. Riedel, É. Gaussier, and G. Bouchard, "Complex embeddings for simple link prediction," in *Proc. ICML*, 2016, pp. 2071-2080.
- [24] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," *arXiv:1810.04805*, 2018.
- [25] M. Mayank, P. Agarwal, and M. Sachan, "DEAP-FAKED: Knowledge graph based approach for fake news detection," in *Proc. IEEE/ACM ASONAM*, 2022.
- [26] L. Cui, D. Lee, and H. Liu, "DETERRENT: Knowledge guided graph attention network for detecting healthcare misinformation," in *Proc. KDD*, 2020, pp. 492-502.
- [27] J. Ma, W. Gao, and K.-F. Wong, "KAPALM: Knowledge graph enhanced language models for fake news detection," in *Proc. EMNLP*, 2023, pp. 3999-4009.
- [28] A. Hogan, E. Blomqvist, M. Cochez, C. d'Amato, G. de Melo, C. Gutierrez, J. E. L. Gayo, S. Kirrane, S. Neumaier, A. Polleres, R. Navigli, A.-C. Ngonga Ngomo, S. M. Rashid, A. Rula, L. Schmelzeisen, J. F. Sequeda, S. Staab, and A. Zimmermann, "Knowledge graphs," *ACM Computing Surveys*, 2021.
- [29] P. Veličković, G. Cucurull, A. Casanova, A. Romero, P. Liò, and Y. Bengio, "Graph Attention Networks," in *Proc. ICLR*, 2018.
- [30] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735-1780, 1997.
- [31] F. Scarselli, M. Gori, A. C. Tsoi, M. Hagenbuchner, and G. Monfardini, "The graph neural network model," *IEEE Transactions on Neural Networks*, vol. 20, no. 1, pp. 61-80, 2009.
- [32] J. Zhou, G. Cui, S. Hu, Z. Zhang, C. Yang, Z. Liu, L. Wang, C. Li, and M. Sun, "Graph neural networks: A review of methods and applications," *AI Open*, vol. 1, pp.